Games on Higher Dimensional Automata

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Abstract. Higher Dimensional Automata are a very powerful model of concurrency to represent truly parallel events, i.e. events that happen simultaneously. They are therefore more expressive than classical automata by allowing non-interleaving concurrency.
In this work, we extend classical notions of games on graphs to Higher Dimensional Automata : games are represented by an arena where several players coexist (some of whom may be allies) with (possibly different) objectives, as accessibility or safety.

Keywords : Higher Dimensional Automata, Games on graphs, Automata

Introduction : Higher Dimensional Automata

Higher Dimensional Automata (HDA) is a geometric model introduced by Pratt in 1991 [10]. This model generalizes previous models of concurrency (e.g. Safe Petri-nets) [9] and automata by allowing non-interleaving concurrency : events can happen simultaneously and not just one after another. The language associated to this model is presented in [5] : HDA are not represented with simple word but with partially-ordered multiset (pomset) with interval order and interface. To represent that event a et b are running in parallel we use event order $\dashrightarrow$: $a \dashrightarrow b$. Respectively, to represent that a has happened before b, we write $a \rightarrow b$. These two partial order complete themselves into a total order. The interval order ensures then that the event history can be decomposed as intervals of time . Interfaces (denoted $\bullet a$ or $a\bullet$) indicate event that was already active or still active at the begin/end of a pomset.

Let us describe an example of Higher Dimension Automata of dimension 2 (as only two events can happen in parallel) in Figure 1. A Higher Dimensional Automata is composed of cells of varying dimension, that represent the number of parallel events. Here in our example, we have cells of dimension 0 (circles), 1 (transition), 2 (grey faces) for cells where respectively 0, 1 or 2 events happen in parallel.

Games on HDA

The first research on Higher Dimensional Automata [10] focused on a geometrical and categorical point of view. For the past few year [1, 2, 3], they have been studied from an automata and verification point of view. In particular, they can define several semantics of numerous concurrent models such as Safe Petri Net [9].

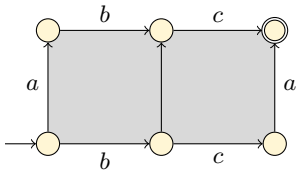

Fig. 1: An example of HDA where a can execute simultaneously with b and c but b and c cannot.

Games on graphs or automata, as presented in [6], allow including multiple players that are in concurrency for the control of an arena. Our aim is to extend that to true concurrency by exploring games on HDAs. In our model, they control a set of events and aim to reach some cells in the HDA or on the contrary to avoid them.

Ongoing work

Our current work consists in understanding the game theoretic notions (game, strategy, outcome...) on an arena modelled as an HDAs. In particular, we describe the unfolding of HDAs suited to represent the strategy of the controller, inspired from the one presented by Dubut in [4].

Our goal is to explore different type of games on Higher Dimensional Automata with multiple players. As a first step, we study zero-sum reachability and safety games and aim to extend the formalism of [7] (originally for Petri Nets) to Higher Dimensional Automata.

In the future, we would like to develop multi-strategies (i.e. where player can propose multiple choices) as presented in [8] in order to introduce a notion of robustness to games on HDAs.

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